

International trade network: statistical analysis and modeling

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Abstract

Globalization has rapidly advanced in recent decades, fostering increased economic interdependence among nations. However, it has also heightened countries' vulnerability to supply chain disruptions, a risk brought into sharp focus during the COVID-19 pandemic. This study provides a comprehensive statistical analysis of bilateral export data for 186 countries across three key years, 2018 (pre-pandemic), 2020 (pandemic peak), and 2022 (post-pandemic recovery), to assess the impact of the pandemic on the structure and dynamics of international trade. We employ Exponential Random Graph Models (ERGMs) to identify the structural and covariate-based determinants of bilateral trade relationships and Stochastic Block Models (SBMs) to uncover latent community structures and clustering behavior within the trade network. Our analysis reveals persistent and statistically significant nodal attributes shaping trade ties, such as GDP per capita, education expenditure, political stability, and access to the sea. Despite expectations of major disruption, the overall structure of the trade network remained remarkably stable throughout the pandemic period, with no substantial changes in network connectivity or class organization. These findings highlight the resilience of the global trade system and contribute to a better understanding of its underlying mechanisms, offering valuable insights for policymakers and researchers in international economics and network analysis.

Keywords: COVID-19, exponential random graph model, international trade, networks, stochastic block model

1. Introduction

Over the past few decades, the world has witnessed a marked process of globalization, resulting in an unprecedented level of interconnection between countries (e.g., Giddens, 2000; Stiglitz, 2002; Friedman, 2005). Globalization has profoundly impacted the dynamics of the global economy, leading to a trend of liberalization, where technological and regulatory developments have promoted the free movement of capital and goods (Piketty, 2014; Rodrik, 2018; World Bank Group, 2020).

Similarly, globalization has transformed global production dynamics, as countries have chosen to specialize in the production of goods where they hold comparative advantages. These goods are then offered in the international market, while the production of items where they lack such advantages is outsourced to other countries. However, despite the benefits of trade dynamics fostered by globalization, this has introduced the risk of disruption in global supply chains. In other words, events that impede the normal flow of commercial activities can lead to shortages of goods and services, with no available alternatives or substitutes (Hansen *et al.*, 2022).

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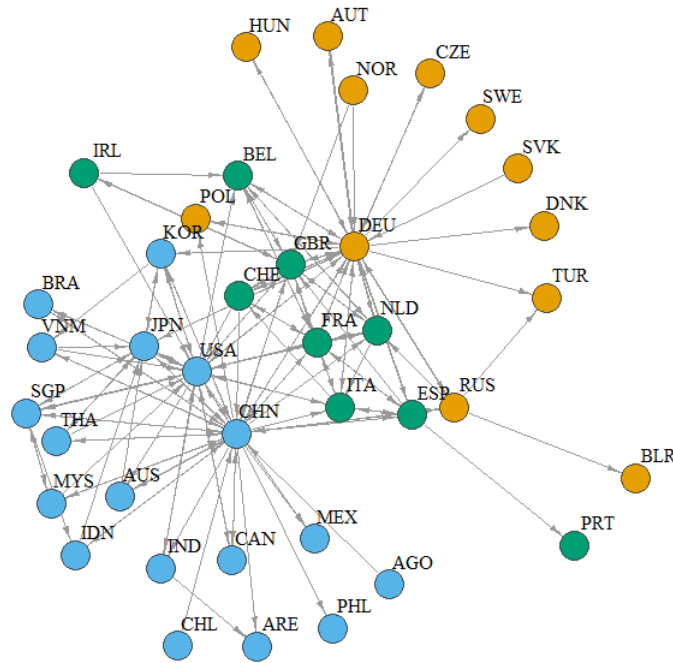


Figure 1: Graph of the international trade network for 2018 filtered by the 150 largest trade relationships. As a preliminary step, the graph was clustered using the Infomap algorithm (Rosvall and Bergstrom, 2008), which initially appears to group countries according to their trade and economical development.

The risk of disruption to global supply chains materialized during the COVID-19 pandemic, a low-probability but high-impact event that disrupted international trade (Petropoulos, 2021). The World Health Organization (WHO) declared COVID-19 a pandemic on March 11, 2020, and it became one of the most profound crises of our time due to its vast socioeconomic impact. To curb the spread of the virus, many countries imposed various restrictions on individuals and businesses, including mandatory lockdowns and bans on foreign entry. According to Hayakawa and Mukunoki (2021), the strictest restrictions occurred during the first wave of the pandemic, which took place in the second quarter of 2020. However, during the second wave in the third quarter of 2020, countries began to gradually lift these restrictions, leading to a slow recovery in growth and a gradual reactivation of global economic activity.

As a result of the severe disruption to global supply chains and the product shortages caused by the COVID-19 pandemic, Hansen *et al.* (2022) argue that several authors began to consider that the pandemic might signal the end of globalization or at least a permanent alteration of the process. To assess whether the COVID-19 pandemic had an impact on globalization processes, it is essential to analyze trade dynamics—from their determinants to their fundamental characteristics—before, during, and after the pandemic. Therefore, this article employs several network analysis methodologies to understand the short- and long-term implications of the pandemic. This approach aligns with Almog *et al.* (2015), who suggest that different economic crises have renewed interest in the study of trade networks, as economic interactions between countries are the primary channels through which

economic shocks are propagated.

Several authors have analyzed a wide range of international trade network configurations to provide decision-making tools and inform policy development. Notable among them is Abbate *et al.* (2012), who studied trade networks in response to geographical and temporal changes, using diverse subnet configurations to demonstrate that the global trade network is disassortative at long distances but assortative at short distances. This contrasts with Silva *et al.* (2024), who found that the network is generally disassortative. Another example is Gutiérrez-Moya *et al.* (2020), who explored the global wheat trade network through an Exponential Random Graph Model (ERGM), incorporating factors such as reciprocity, economic openness, country size, and land area. Their findings suggest that these factors are crucial in establishing trade links between countries. Additionally, Amin *et al.* (2022) analyzed the global trade network for the years 2011, 2013, 2015, and 2017, using a broad range of ERGM configurations and characteristics. They concluded that positive effects of reciprocity and transitivity prevail, alongside the influence of non-structural factors in the formation of trade links.

It is also worth noting that the most popular methodologies used to study bilateral trade relations include: Gravity Equation Models for predicting links (e.g., Eaton and Kortum, 2002; Silva and Tenreyro, 2006; Baier and Bergstrand, 2009), as employed by Almog *et al.* (2015), Satya and Vinicios (2023), and Hayakawa and Mukunoki (2021); Fitness Models to replicate network topology (e.g., Bianconi and Barabási, 2001; Caldarelli and Vespignani, 2007; Miller and Jackson, 2008), as presented by Almog *et al.* (2015); and ERGMs to identify the determinants of bilateral trade relations (e.g., Frank and Strauss, 1986; Wasserman and Pattison, 1996; Robins *et al.*, 2007), as carried out by Amin *et al.* (2022) and Gutiérrez-Moya *et al.* (2020).

This article characterizes the international trade network, from defining its topological properties to identifying the determinants of bilateral trade relations, as well as examining the role of different countries within the network. Additionally, it determines whether the COVID-19 pandemic led to short- or long-term changes in the network's structure. With this in mind, the study employs various methodologies for analyzing relational data, such as calculating and interpreting the structural characteristics of the network, using ERGMs (Hunter and Handcock 2006; Snijders *et al.* 2010; Lospinosa and Fienberg 2021) as well as Stochastic Block Models (SBMs) (Holland *et al.*, 1983; Nowicki and Snijders, 2001; Karrer and Newman, 2011; Peixoto, 2014; Fortunato and Hric, 2016; Karrer and Newman, 2017).

This article is structured as follows: the first section presents the data used for the analysis. The second section outlines the theoretical framework behind the methodologies and models employed in the study. The third section discusses the results of the analysis. Finally, the fourth section discusses the main findings and several lines for future research.

2. International trade network: construction and processing

We constructed the international trade network using two primary databases: one containing bilateral trade relations and the other detailing macroeconomic, geographic, and political characteristics of the countries under evaluation. In order to establish bilateral trade relations, we use data on the value of bilateral exports and imports reported annually in thousands of US dollars by the United Nations Statistics Division (UNSD). These data are stored in the World Integrated Trade Solution (WITS) repository. Data from the years 2018, 2020, and 2022 were extracted for 240 territories.

Sajedianfard *et al.* (2021) argue that, in practice, measurements of exports and imports do not necessarily align due to methodological differences in data collection, which typically results in discrepancies. Therefore, export and import data should be analyzed separately. Consequently, in this

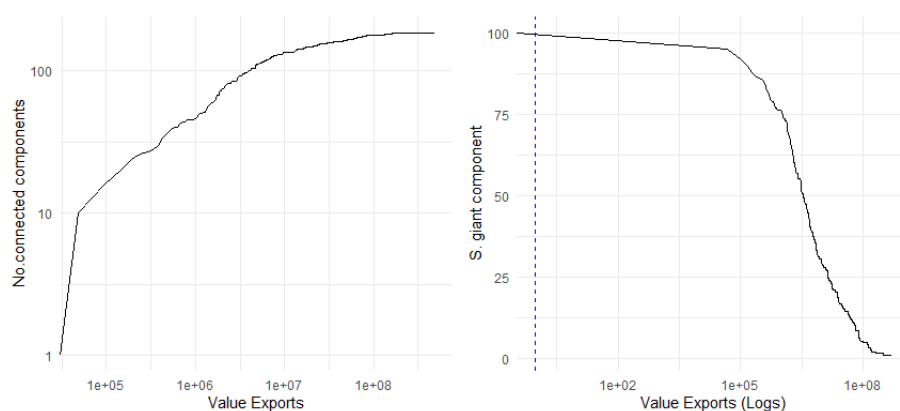


Figure 2: *Connectivity of the 2018 international trade network conditioned by export values. The blue dashed line marks the threshold beyond which the size of the giant component begins to decrease.*

Table 1: Descriptive statistics of the bilateral trade relations database

Metric	2018	2020	2022
Number of countries	186	186	186
Number of observations	16,357	18,017	17,276
Average trade flow	905,674	764,221	1,114,961
Median trade flow	7,054	5,482	8,052
Standard deviation	7,415,086	6,532,794	9,699,058
Minimum	0	0	0
Maximum	479,278,747	452,492,876	582,756,110

All units, except for number of countries and number of observations are expressed in thousands of USD.

article we exclusively focus on export data, primarily concentrating on exporting countries. This approach does not undermine the analysis, as export data should effectively represent the majority of global trade flows, since theoretically, exports from country A to country B should be equal to the imports of country B from country A (Bussière and Chinn, 2018).

The bilateral trade relations database consists of 240 territories, including countries, special administrative regions, economic and monetary unions, among others. For this analysis, we chose countries as the unit of analysis. Therefore, the list of ISO country codes recognized by the United Nations (UN) can be used to filter the original database accordingly. It is important to note that the WITS database does not provide reports for all countries, resulting in a final dataset comprising 186 countries. Table 1 presents the main descriptive statistics of the bilateral trade relations database. The data exhibit considerable dispersion, suggesting significant asymmetries between countries, as evidenced by the difference in magnitude between the median and mean of export data, as well as the elevated magnitude of the standard deviation.

In order to construct the node attribute database, we use macroeconomic, geographic, and political data for the countries under analysis. This information was sourced from the World Bank Development Indicators database (WB) and the Centre d'Études Prospectives et d'Informations Internationales (CEPII). The selection of variables was guided by economic theories concerning potential determinants of international trade.

Table 2 presents the main descriptive statistics of selected variables from the nodal attributes database. Each of the chosen nodal attributes represents a descriptive component of the character-

Table 2: Descriptive statistics of the node attribute database for 2018

Attribute	Mean	SD	Min.	Median	Max.
Agriculture (GDP %)	10.33	10.34	0.02	6.73	58.93
Industry (GDP %)	26.12	12.01	4.87	24.65	65.88
Services (GDP %)	55.87	11.99	32.35	55.54	86.85
GDP per capita (USD)	16,236.00	27,658.80	232.10	6,316.5	193,968.10
GCF (GDP %)	24.61	7.94	2.78	23.35	54.02
Foreign investment (GDP %)	-5.27	61.66	-13.03	2.73	29.21
Inflation (annual %)	4.96	14.21	-2.89	2.79	55.97
Political stability	-0.06	0.97	-2.75	-0.01	1.58
Education expenditure (GDP %)	4.37	1.84	0.23	4.37	15.38
Total population	38.30	150.49	0.01	7.86	1,403
Unemployment (workforce %)	7.39	5.87	0.11	5.20	26.98
Landlocked	0.19	0.39	0	0	1

GDP and GCF stand for Gross Domestic Product and Gross Capital Formation, respectively. Political stability is given in units of a standard normal distribution (ranging from approximately -2.5 to 2.5). Total Population is given in millions. Landlocked is a dummy variable assuming the value 1 if the country does not have access to the sea, and 0 otherwise.

istics of each country, ranging from the definition of comparative advantages to the description of political, economic, and social conditions. The variables ‘Agriculture’, ‘Industry’, and ‘Services’ are selected as proxies for the level of productive specialization in an economy. The variable ‘Gross Domestic Product per capita’ is chosen as a proxy for the level of wealth, and consequently, the potential consumption capacity of an economy. The variable ‘Education Expenditure’ is used as a proxy for human capital. The variables ‘Gross Capital Formation’ and ‘Foreign Investment’ are selected as proxies for infrastructure, technological advancement, and foreign investment. Lastly, the variables ‘Unemployment’, ‘Population’, ‘Inflation’, and ‘Political Stability’ are selected as proxies for the political and social situation of the economies.

2.1. Handling missing data

As in many studies involving macroeconomic data, the collected dataset contains gaps or missing values. In this case, we only consider variables with less than 30% of missing data. To address this issue, the missing data values are imputed using the k -nearest neighbors method (KNN Imputer; Buuren and Groothuis-Oudshoorn 2010; Pedregosa *et al.* 2011). For each missing value in the dataset, the KNN algorithm locates the k nearest neighbors based on Euclidean distance and imputes the missing values by leveraging the observed values of the most similar data points within the feature space (Kowarik and Templ, 2016). We use a cross-validation approach to determine the value of k that minimizes the mean squared error, which give us an optimal number of $k = 1$ neighbors.

2.2. Network consolidation

We define the international trade network a graph generated from bilateral export data between countries obtained from the World Integrated Trade Solution (WITS) for a given time period t . Specifically, the graph G_t is defined as $G_t = (V_t, L_t)$, where V_t represents the set of countries (nodes or vertices) and L_t denotes the number of directed trade links (edges or links), forming the network $N_t = (V_t, L_t, P_t, W_t)$. Here, P_t is the node attribute matrix and W_t is the matrix of link weights. This implies that the links have varying intensities based on the value of trade flows (Abbate *et al.*, 2012).

In general, each pair of nodes (i, j) is referred to as a dyad. For each dyad, the presence or absence of a link is indicated using the adjacency matrix, denoted as Y . As mentioned above, the network is directed; thus, for any two nodes i and j , $Y_{i,j} = 1$ ($Y_{i,j} = 0$) represents the presence (absence) of a

directed edge from node i to node j . Specifically, $Y_{ij} = 1$ indicates that there is a trade relationship where country i exports to country j . Additionally, the weight represents the magnitude of the exports, measured in thousands of US dollars. Since the network is directed, Y is not necessarily symmetric (Lee and Wilkinson, 2019). Figure 1 shows the graph for the 2018 international trade network filtered by the most notable countries of each class.

3. Methodology

In this section, we present the key methodological aspects for studying international trade networks. We primarily focus on a detailed characterization of ERGMs and SBMs. For a gentle introduction to these topics, we recommend consulting the book by Kolaczyk and Csárdi (2020).

ERGMs were applied to capture global network structures and understand the overall patterns of connectivity in the international trade network. By modeling the presence or absence of trade relationships as a function of structural and covariate-driven factors, ERGMs allowed us to evaluate key determinants of network formation, such as reciprocity, transitivity, and economic attributes. This global perspective was essential for addressing our research question regarding the structural impacts of the COVID-19 pandemic on the trade network across distinct phases.

In contrast, SBMs were used to identify local patterns and detect latent clusters of countries within the network, reflecting group-level behaviors and structural tendencies. This approach provided insights into the community organization of the trade network and how these clusters evolved or remained stable across the pre-pandemic, pandemic, and recovery periods. The use of SBMs complemented the global insights from ERGMs by uncovering meso-level structures, such as cohesive trading blocs or regional groupings, that are not easily captured by ERGMs.

Together, these approaches provided a holistic framework for understanding the trade network's structure. ERGMs offered a global, pattern-focused analysis, while SBMs revealed localized, cluster-based dynamics. This dual approach allowed us to comprehensively address our research objectives by capturing both the macro-level and meso-level dimensions of the network.

3.1. ERGM

ERGMs serve as a direct extension of classical generalized linear models (GLMs) (McCullagh and Nelder, 1989; Faraway, 2016; Hastie *et al.*, 2017) to the context of network data. The formulation of ERGMs is intended to adapt and extend the principles and methods of GLMs to the unique characteristics of network data. While GLMs are used to model relationships between variables in traditional statistical settings, ERGMs are specifically designed to analyze the structure of networks by modeling the probability of observing a particular network configuration.

ERGMs, also known as p -star models, allow us to analyze link formation within networks. A key distinction from regression models is their ability to account for the interdependence of network links, relaxing one of the most restrictive assumptions inherent in regression models. This capability allows ERGMs to describe the fundamental forces shaping the global network structure. In this way, ERGMs enable the inclusion and consideration of node characteristics and structural configurations (Amin *et al.*, 2022). This makes them particularly suited for capturing the dependency relationships that define the global structure of international trade networks. That is why we use ERGMs to identify the principal factors influencing the formation of trade relationships between countries.

The general form of ERGMs specifies the probability of the entire network as a function of a set of relevant features. Consider the random graph $G = (V, E)$, where $Y_{i,j}$ is a binary random variable indicating the presence or absence of an edge $e \in E$ between two vertices i and j in V . The basic

specification for an ERGM is as follows,

$$\Pr(Y = y) = \frac{1}{\kappa} \exp(\theta_1 g_1(y) + \theta_2 g_2(y) + \cdots + \theta_p g_p(y)),$$

where:

- Y is the adjacency matrix.
- κ is the normalization constant.
- $g_h(y)$ takes the value 1 if the edge configuration h is present in y , and 0 otherwise.
- θ_h is an unknown parameter indicating that dyads are dependent in the edge configuration h .

It is important to note that, given a choice of functions g_H and their coefficients θ_H , there is a corresponding structure of dependence among the elements of Y . In general terms, this structure can be described as specifying that the random variables $\{Y_{ij}\}_{(i,j) \in A}$ are independent of $\{Y_{i'j'}\}_{(i',j') \in B}$, conditional on the values of $\{Y_{i''j''}\}_{(i'',j'') \in C}$, for some given index sets A , B , and C (Kolaczyk and Csárdi, 2020).

The ERGMs framework is highly adaptable, offering various modifications and extensions suitable for different types of network data. It can be applied to both directed networks (such as trade networks) and undirected networks. Additionally, ERGMs enable the incorporation of vertex-specific information (such as economic indicators in trade networks), which enhances the model's ability to capture node-level heterogeneity. When applied to the global trade network, the ERGM framework aims to identify significant parameters that accurately represent the network's structure and relationships. By fitting the model to the trade data and analyzing these parameters, we are able to uncover underlying patterns and factors that influence trade interactions between countries, while providing valuable insights into the complexities of the global trade system.

In this study, we use the **statnet** package (Handcock *et al.*, 2018) in R (R Core Team, 2023) to fit the ERGM. **statnet** offers comprehensive tools for network analysis, allowing us to specify and estimate ERGMs effectively.

In this study, the ERGMs were fitted using the **statnet** suite of packages in R (Handcock *et al.*, 2018), which provides robust tools for estimating and analyzing network models. The estimation was performed using Markov Chain Monte Carlo Maximum Likelihood Estimation (MC-MLE), a well-established computational method for ERGMs. This approach iteratively approximates the likelihood of the network model through a combination of Gibbs sampling and iterative reweighting, ensuring reliable estimation for complex network configurations.

For this analysis, we relied on the default settings provided by **statnet**, which are optimized for a broad range of network data. Specifically:

- The number of burn-in steps and MCMC sample sizes were set to the defaults, balancing computational efficiency and model precision.
- Pseudolikelihood-based starting points were used to improve convergence and estimation stability.
- Convergence diagnostics, such as MCMC trace plots and Geweke statistics, were employed to verify the adequacy of the algorithm's convergence.
- Post-estimation diagnostics, including goodness-of-fit tests and visual comparisons of simulated networks to observed data, validated the adequacy of the fitted models.

3.2. SBM

SBMs are increasingly popular in statistical network analysis because they allow for the identification of latent group structures within a network (Lee and Wilkinson, 2019). By grouping nodes into blocks or clusters based on shared connection patterns, SBMs offer a powerful framework for detecting underlying communities or clusters. This approach is especially useful for simplifying complex networks and improving the efficiency of clustering tasks, as it reduces noise by focusing on the most meaningful interactions.

Following very closely Lee and Wilkinson (2019) and Kolaczyk and Csárdi (2020), in SBMs, each node is assigned to one of $K < n$ unknown groups or clusters, where n is the number of nodes. For each node i , we define Z_i as a K -dimensional vector such that

$$z_{i,p} = \begin{cases} 1, & \text{if node } i \text{ belongs to community } p; \\ 0, & \text{otherwise.} \end{cases}$$

By employing these K -dimensional vectors, we can construct a $K \times K$ matrix C in such a way that each entry $C_{p,q}$ represents the probability of a directed edge originating from a node in group p and terminating at a node in group q . This matrix C is crucial for understanding and modeling how edges are formed within the network based on the group affiliations of the nodes. Specifically, C captures the likelihood of interactions between different groups, thereby providing insight into the underlying mechanisms of edge formation within the network. By analyzing this matrix, we can infer how the network's structure is influenced by the block assignments and better understand the dynamics of connections across different groups.

Specifically, given Z and C , the model can be expressed under the assumption that edges are distributed according to a Bernoulli distribution conditional on group memberships. For directed networks, the likelihood is given by

$$p(Y | Z, C) = \prod_{i,j:i \neq j} (Z_i^T C Z_j)^{Y_{i,j}} (1 - Z_i^T C Z_j)^{1-Y_{i,j}}.$$

However, in practical applications with real data, Z and C are not known (as well as the number of groups), and thus, must be inferred. To address this issue, we assume that $\Pr(Z_{i,p} = 1 | \theta_p) = \theta_p$, where $\sum_{p=1}^K \theta_p = 1$. Therefore, given the values of $Z_1, \dots, Z_n$, dyads can be modeled as conditionally independent according to the Bernoulli distribution:

$$y_{i,j} | \mathbf{C}, Z_i, Z_j \sim \text{Bernoulli}(C_{\xi_i, \xi_j}),$$

where $\xi_i = \xi(Z_i)$ denotes the position in Z_i with a non-zero entry (i.e., $\xi_i = p$ indicates that node i belongs to community p).

Kolaczyk and Csárdi (2020) emphasize that estimating the SBM using maximum likelihood methods involves substantial computational costs, particularly due to the complexity of calculating the exact likelihood function for large networks. This computational burden makes it challenging to apply traditional methods, especially in cases with large-scale networks. To address these challenges, the authors suggest using the variational Expectation-Maximization (EM) (Beal, 2003; Wainwright and Jordan, 2008; Blei *et al.*, 2017) algorithm as a more efficient alternative for model fitting. This algorithm circumvents the need to compute the full likelihood by optimizing a lower bound on the likelihood of the observed data, commonly referred to as the Evidence Lower Bound (ELBO). By doing so, the variational EM approach reduces the computational load and enables the estimation

of the SBM in a more scalable and practical manner, while still maintaining reasonable accuracy in capturing the underlying community structure of the network.

In this study, the SBMs were fitted using the **blockmodels** package in R Matias Léger (2016), which implements the variational Expectation-Maximization (VEM) algorithm. This approach is particularly suitable for analyzing large-scale networks, as it efficiently estimates model parameters by optimizing the Evidence Lower Bound (ELBO) rather than directly computing the full likelihood. This reduces computational complexity while maintaining a high degree of accuracy in uncovering latent group structures within the network.

The VEM algorithm combines variational inference with Expectation-Maximization (EM) principles. Specifically:

- **Initialization:** Model parameters are initialized using random or heuristic starting points to ensure convergence to a local optimum.
- **E-Step:** Variational inference estimates the posterior probabilities of node assignments to latent blocks, utilizing the observed network structure.
- **M-Step:** The algorithm maximizes the expected complete-data log-likelihood to update the block interaction matrix and class probabilities.
- **Convergence:** The algorithm iterates until predefined convergence criteria, such as stabilization of the ELBO, are satisfied.

For this analysis, we relied on the default settings of the **blockmodels** package, which are designed to handle a variety of network types. To validate the model fit, post-estimation diagnostics, such as clustering evaluation and visualization of the resulting block structure, were performed. The optimal number of blocks was determined using the integrated classification likelihood (ICL) criterion, which balances interpretability with model fit.

4. Results

In this section, we present our key findings, including a comprehensive characterization of international trade networks for 2018, 2020, and 2022 using structural descriptive analysis, as well as full modeling analysis using ERGMs and SBMs. All results are fully reproducible, and the corresponding R code is openly available at <https://github.com/aarevaloa/Red-de-comercio-mundial>, ensuring transparency and enabling replication of the analysis.

4.1. Network structure

Table 3 displays the main structural characteristics of the international trade networks for 2018, 2020, and 2022. From 2018 to 2020, the number of edges increased by 1,660 connections, while from 2020 to 2022, the number of edges decreased by 741. This suggests a shift in long-term economic dynamics in 2020, followed by a return to previous trends in 2022. This pattern is consistent with observations during the COVID-19 pandemic, where a significant acceleration in e-commerce growth was driven by strong export performance in East Asian economies (UNCTAD, 2021).

Although the pandemic led to a decrease in export values, this did not correspond to a reduction in the number edges. The economic, social, and health crises caused by COVID-19 initially did not result in the destruction of trade relationships but rather in a weakening of trade values, as reflected by the decline in the average edge weights in 2020. This weakening of trade ties prompted the formation of

Table 3: Structural statistics of the international trade network

Statistic	2018	2020	2022
Nodes (n°)	186	186	186
Edges (n°)	16.357	18.017	17.276
Components (n°)	1.00	1.00	1.00
Mean out-degree	87.94	96.86	92.88
CV out-degree	85.01	73.35	78.97
Mean in-degree	87.94	96.86	92.88
CV in-degree	23.80	25.43	24.65
Mean weight	905.67	764.22	1,114.96
CV weight	818.73	854.83	869.90
Degree correlation	0.66	0.72	0.74
Density	0.47	0.52	0.50
Transitivity	0.76	0.79	0.78
Assortativity	-0.23	-0.24	-0.24

CV: Coefficient of variation (%). Weight in millions of USD.

new trade relationships among agents who had not previously traded with each other, as part of efforts to secure essential supplies and food to address the health crisis and the needs of the population. By 2022, we observe a recovery in export values, as indicated by the increase in the average edge weights, with levels surpassing those seen in the pre-pandemic period. Simultaneously, there is a reduction in the number of trade relationships, which can be attributed to the stabilization of production structures across economies.

There is a clear presence of a disassortative relations. Specifically, the assortativity coefficients for 2018, 2020, and 2022 are -0.23 , -0.24 , and -0.24 , respectively. These negative values indicate a persistent trend where highly connected nodes tend to form associations with nodes that have substantially fewer connections. Rather than forming clusters with similarly connected nodes, central or highly networked nodes preferentially interact with less connected, peripheral nodes. This finding aligns with prior research (e.g., Abbate *et al.*, 2012; Silva *et al.*, 2024), which also identified a marked disassortative pattern in commercial relationships within the international trade network. Such a pattern suggests a distinctive structural dynamic in global trade, where key players engage with smaller or less prominent countries, reinforcing the hierarchical nature of these interactions. Such disassortativity has implications for understanding the flow of goods, influence, and resilience within the network.

The coefficients of variation exhibit consistently high values, which corresponds to a pronounced heterogeneity in how countries are integrated into the international trade network. This substantial variability suggests that countries do not participate in global trade on equal terms; instead, nations assume predefined roles within the network of trade relations. These roles may be influenced by various factors such as economic size, geopolitical position, trade agreements, or the nature of their export and import markets. The elevated coefficients underscore the unequal distribution of trade connections, where some countries act as central hubs with extensive trade links, while others maintain more peripheral roles with fewer connections. This structural disparity highlights the complexity exhibited by global trade, where the dynamics of influence, access to markets, and economic interdependence vary greatly from one country to another.

The international trade network is characterized by a high level of connectivity. This is evident from the network's density, which hovers around 0.5 for all three years. Such high connectivity is a key indicator of the network's global cohesion and it constitutes a likely explanation for the network's transitivity. The underlying reason for this pattern can often be found in the dynamics of trade treaties

and the creation of regional trade blocs. Once two countries establish a commercial relationship, they are more likely to enter into multilateral agreements or participate in regional trade organizations that encourage broader interaction. As such, the high density of the network reflects not only the individual trade agreements between pairs of countries but also the broader institutional structures that facilitate global trade, such as free trade agreements, customs unions, and economic partnerships.

Finally, the analysis shows that the network remains a single connected component across all three years. This indicates that, despite geopolitical tensions and geographic barriers, global trade is effectively maintained, allowing for the exchange of goods and services between any pair of countries. The presence of a single component highlights the global nature of trade, where no country is fully isolated from the system. Despite regional conflicts, trade disputes, or other obstacles that may create friction in international relations, these findings suggest that such challenges do not fragment the overall network. Instead, countries remain interconnected through various trade routes, often facilitated by intermediaries or regional trade agreements, ensuring that global commerce continues to operate as a cohesive system.

4.2. Network connectivity

To explore the connectivity of the global trade network, we follow very closely the reasoning provided in Abbate *et al.* (2012). Specifically, we partition the network based on net export values to create subgraphs with thresholds of export values. This approach allows us to analyze how the network's connectivity evolves as we introduce higher-weight edges. In this way, the changes in connectivity were assessed through metrics such as the number of connected components and the size of the largest connected component.

Figure 2 illustrates how connectivity statistics change with accumulated export values for 2018 (similar connectivity graphs for 2020 and 2022 are not shown here since they are quite similar). Generally, as higher-value trade relationships are added to the network, the number of connected components increases. Initially, when only trade relationships with the lowest export values are included, the network shows a rapid rise in the number of connected components. As trade relationships with higher export values are progressively added, the network's connectivity improves, leading to an increased number of connected components, enhancing the integration and cohesion of the network.

The network remains cohesive primarily due to numerous lower-value connections, which are crucial for maintaining overall connectivity. When these weaker connections are removed, the network fragments rapidly, revealing that only a few countries with significant high-value exports form its core. These core hubs anchor the network, while its stability depends on the extensive web of lower-value connections. The stabilization at higher values after the removal of these weaker links indicates that the remaining network comprises several smaller, densely interconnected groups of countries. This fragmentation also highlights that many nations outside the core rely heavily on these lower-value connections for their integration, becoming isolated when these links are removed.

The size of the giant component as a function of the logarithm of export values illustrate how the giant component's size changes when only higher-value edges are considered. Initially, the giant component encompasses 100% of the nodes, as all nodes are connected within a single component. However, as lower-value connections are removed, the size of the giant component begins to decrease but remains relatively stable until reaching a certain threshold (the inflection point of the graph). This indicates that the size of the giant component is primarily dependent on higher-value edges.

Our findings reflect the characteristics of the international trade network, where high-value connections are linked to large economies that export substantial quantities of goods and services, which are relatively rare due to the limited number of such economies. These large economies form cen-

Table 4: ERGM results for the 2018 network

Variable	Estimation	SD	z value	Significance
edges	-3.32	0.35	-9.493	***
nodecov agriculture	-0.040	0.002	-20.104	***
nodecov industry	0.010	0.002	5.127	***
nodecov foreign investment	-0.001	0.000	-4.815	***
nodecov PIB per capita	0.000	0.000	-26.433	***
nodecov education expenditure	-0.081	0.006	14.177	***
nodecov GCF	0.021	0.001	16.781	***
nodecov inflation	0.010	0.001	8.043	***
nodecov political stability	0.070	0.010	5.522	***
nodecov total population	0.000	0.000	7.733	***
nodecov services	0.030	0.002	13.270	***
nodecov unemployment	0.001	0.002	0.783	
nodematch agriculture	0.080	0.500	0.153	
nodematch industry	-1.030	1.050	-0.977	
nodematch foreign investment	-0.230	0.550	-0.425	
nodematch PIB per capita	-10.010	70.020	-0.143	
nodematch education expenditure	0.510	0.240	2.080	*
nodematch GCF	-0.370	0.300	-1.221	
nodematch inflation	-2.320	1.090	-2.140	*
nodematch political stability	1.340	1.080	1.241	
nodematch total population	9.380	70.020	0.134	
nodematch services	1.220	1.230	0.991	
nodematch unemployment	-0.450	0.370	-1.219	
nodefactor landlocked	-0.300	0.020	-13.736	***

Edges: Total number of edges in the graph. Nodecov: Sum of the attributes. Absdiff: Absolute difference of the attributes. Nodematch: Number of attribute matches. Significance codes: p -value < 0.1 (·), p -value < 0.05 (*), p -value < 0.01 (**), p -value < 0.001 (***).

tral hubs in the network due to their extensive trade relationships. In contrast, smaller economies, with limited export capacity, rely on a few low-value connections and trade with fewer countries. The removal of these lower-value connections causes these smaller economies to gradually detach from the giant component. Conversely, when higher-value edges are removed, the central hubs (large economies) begin to disconnect, leading to the rapid disintegration of the giant component, highlighting the crucial role of these large economies in maintaining network cohesion.

4.3. ERGM

This section presents the results regarding the ERGM, considering key non-structural characteristics influencing the international trade network. Notably, we identify several statistically significant effects on the formation of network edges. Table 4 shows the results for the 2018 network (similar tables for 2020 and 2022 are not shown here since they are quite similar).

Note that the covariates ‘nodecov agriculture’, ‘nodecov foreign investment’, ‘nodecov GDP per capita’, and ‘nodecov education expenditure’ are statistically significant with negative estimates. This result indicates that the probability of two countries establishing trade relations decreases as the presence of one or more of these characteristics weakens within their economies. Additionally, these four characteristics are established as fundamental descriptors of a country’s economic structure. Consequently, as two countries display divergent economic conditions, the likelihood of trade between them is expected to decline. This finding aligns with Frankel and Kenan Rose (2001), which reported that dissimilar economic characteristics between countries adversely affect the formation of trade relationships. Additionally, we observe the previous behavior persists over time. For both 2020 and 2022, the

Table 5: Significance of the estimates: 2018, 2020, and 2022

Variable	2018	2020	2022
edges	***		***
nodecov agriculture	***	***	***
nodecov industry	***	***	***
nodecov foreign investment	***	***	
nodecov PIB per capita	***	***	***
nodecov education expenditure	***	***	**
nodecov GCF	***	***	***
nodecov inflation	***		***
nodecov political stability	***	***	***
nodecov total population	***	***	***
nodecov services	***	***	***
nodecov unemployment		***	***
nodematch agriculture		*	
nodematch industry		.	
nodematch foreign investment			
nodematch PIB per capita			
nodematch education expenditure	*	*	
nodematch GCF		**	
nodematch inflation	*	**	***
nodematch political stability			
nodematch total population			
nodematch services			
nodematch unemployment			.
nodefactor landlocked	***	***	***

Significance codes: p -value < 0.1 (.), p -value < 0.05 (*), p -value < 0.01 (**), p -value < 0.001 (***).

same covariates consistently show negative and statistically significant estimates.

The covariates ‘nodecov industry’, ‘nodecov GCF’, ‘nodecov inflation’, ‘nodecov political stability’, ‘nodecov total population’, and ‘nodecov services’ exhibit positive and statistically significant estimates, suggesting that these factors are associated with an increased likelihood of establishing trade links. Independent evaluation of these covariates is consistent with economic literature. For example, ‘nodecov inflation’ indicates that rising inflation often results in currency depreciation as agents move their assets to safer investments. This depreciation makes local goods cheaper internationally, increasing export demand. However, the effectiveness of raising interest rates to combat inflation depends on perfect arbitrage between countries; otherwise, its impact will vary based on factors such as expectations, institutional confidence, and foreign exchange market elasticity.

The covariate ‘nodecov inflation’ is statistically significant in 2018 but it is not in 2020 (see Table 5). This shift can be attributed to the pandemic-induced sharp decline in household and business consumption, which suppressed inflationary pressures and led to a global period of low inflation. As a result, the variable loses its significance as a determinant of trade between countries during this period. However, by 2022, inflation levels had reverted to pre-pandemic values due to economic recovery and increased consumption by households and businesses, which corresponds with the variable’s return to statistical significance.

When evaluating nodal characteristics by differences, only ‘nodematch education’ expenditure’ and ‘nodematch inflation’ are significant. The former, with a positive estimate, suggests that countries with similar educational spending are more likely to trade with each other, while the latter, with a negative estimate, indicates that countries with similar inflation levels are less likely to trade. Additionally, in 2020, three other nodal characteristics are significant with positive coefficients, namely, ‘nodematch agriculture’, ‘nodematch industry’, and ‘nodematch GCF’. This suggests that countries

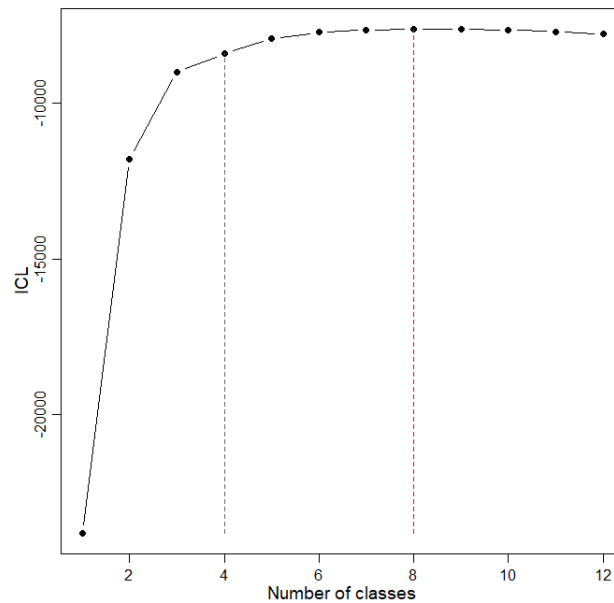


Figure 3: Selection of the number of classes using the ICL criterion.

with similar profiles in these areas are also more inclined to trade. These findings reflect increased investment in healthcare and research during the COVID-19 crisis and a shift toward agriculture and industry due to service sector slowdowns.

The nodal characteristic ‘nodecov unemployment’ becomes statistically significant in 2020 with a positive estimate, indicating that the presence of this characteristic increases the likelihood of trade between countries. This finding is consistent with observations during the COVID-19 crisis, where mobility restrictions negatively impacted production and, according to Okun’s Law (a direct relationship between production and employment levels; Okun 1962), increased unemployment. This situation may have compelled economies to boost trade to secure essential goods. Finally, the nodal characteristic ‘nodefactor landlocked’ (access to the sea) reveals a negative and statistically significant relationship with the formation of trade links. This suggests that landlocked countries are less likely to engage in international trade.

4.4. SBM

In order to understand the structure of the trade network and cluster countries according to their structural characteristics, we implement here a SBM for each year. The results for all three time periods are nearly identical, both in terms of the number of optimal classes and the resulting groupings. Therefore, we present the results only for the year 2018, as the conclusions and results are consistent across the other periods. This suggests that there was no substantial change in the structure of the international trade network during or after the pandemic.

The number of classes and their assignments are typically unknown in real-world data. Here, we use the integrated classification likelihood (ICL) criterion to identify the number of classes. Higher ICL values indicate a better model fit. We test class numbers Q from 1 to 12. Figure 3 shows that the ICL increases with the initial classes and stabilizes at $Q = 8$ (see dashed red line). According to Ko-

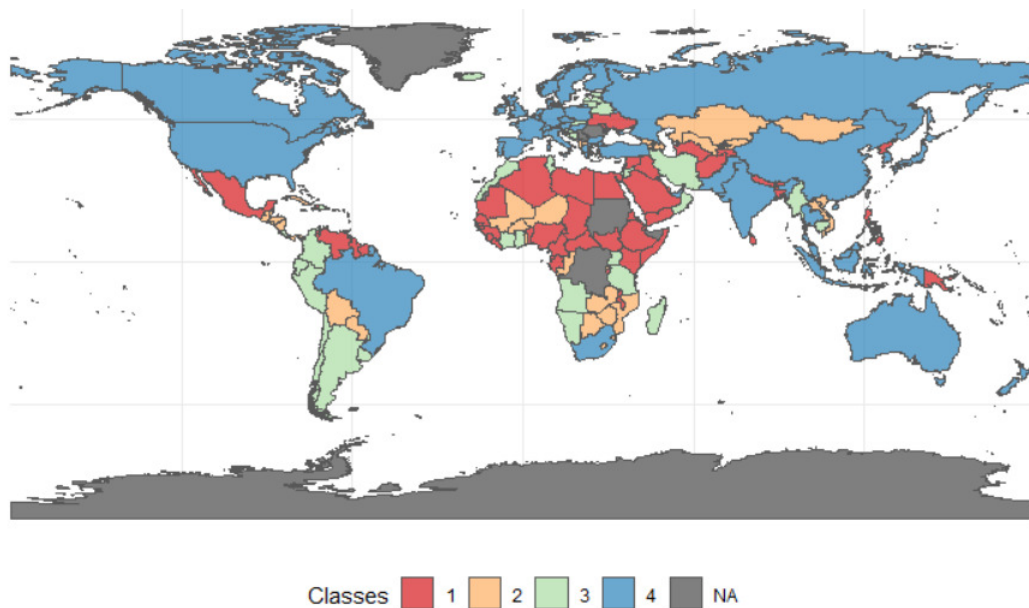


Figure 4: Clustering of countries based on their structural characteristics in the 2018 international trade network, using a SBM.

laczyk and Csárdi (2020), while the ICL helps determine the class number, economic meaning favors a lower number of classes. Thus, we select $Q = 4$ to maintain interpretability without significantly reducing the ICL.

Now, we fit the SBM using $Q = 4$ classes. Results are presented in Figure 4. Class 1 (79 countries) includes countries such as Vietnam and Azerbaijan; Class 2 (39 countries) includes countries such as Yemen and Palau; Class 3 (39 countries) includes countries such as Slovakia and Chile; and Class 4 (38 countries) includes countries such as China and Germany. Since the model considers only relational structural characteristics of countries, the resulting clustering reflects commercial development. Generally, countries in Class 1 exhibit less commercial development than those in Class 2, countries in Class 2 exhibit less than those in Class 3, and countries in Class 3 exhibit less than those in Class 4.

Figure 5 shows the international trade network segmented those classes using the SBM, focusing on the six countries with the highest export value added per class. In the figure, vertex colors represent country classes, and edge colors correspond to the exporting country. As established in Section 4.2, the international trade network includes many countries with low-value connections, while a few generate high-value connections due to their larger economies. The predominance of blue edges, linked to the most trade-capable countries, shows that Class 4 countries serve as central hubs. The number of edges decreases for countries in classes 3, 2, and 1, reflecting their lower export capacity.

Figure 6 presents the adjacency matrix ordered by country classes. Note that countries in class 1 exhibit a vague structure compared to the countries in the other classes. This is due to the fact that countries belonging to class 1 are those with the lowest level of development and, therefore, maintain significantly fewer trade relationships than those countries in the other classes. However, there are no further patterns of relationships within or between classes, which is expected given current trade dynamics. Unlike other networks, where within-class ties are stronger, international trade involves

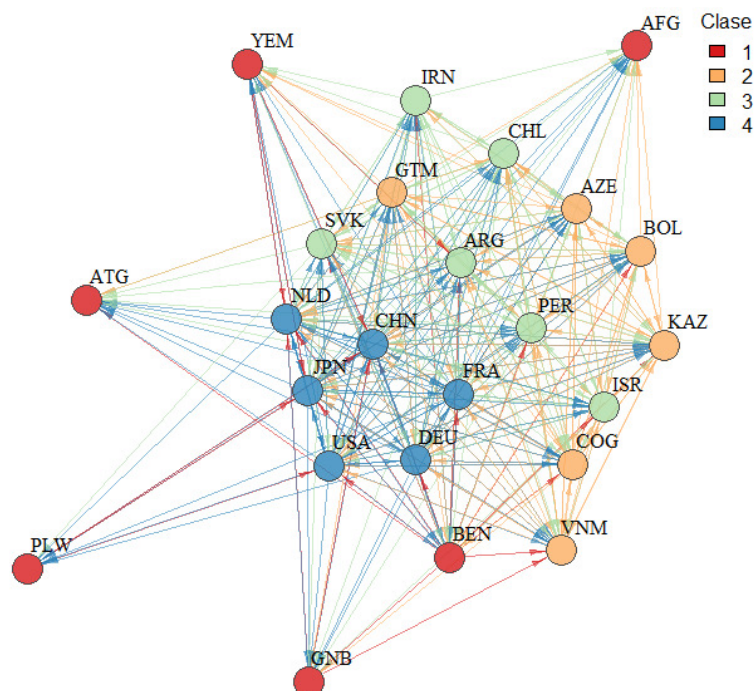


Figure 5: Graph of the international trade network for 2018, differentiated by country classes and filtered to show the six largest exporters in each class. Vertex colors represent country classes, and edge colors correspond to the exporting country.

countries trading with both similar and different partners. This is due to the international division of labor, where countries specialize in goods with comparative advantages and trade for those they cannot produce as efficiently.

5. Discussion

Both the network structure analysis and the ERGM identified a consistent set of non-structural node characteristics as significant determinants of international trade across the years 2018, 2020, and 2022. These include variables such as agriculture, industry, services, GDP per capita, investment, education expenditure, inflation, and access to a coastline. During the pandemic, the ERGM results revealed a shift in key factors, with unemployment, differences in agricultural and industrial output, and gross capital formation becoming significant. These changes reflect the immediate economic disruptions caused by the pandemic. The findings align with prior studies, including Amin *et al.* (2022) and Abbate *et al.* (2012), which also emphasize the role of these macroeconomic factors in shaping international trade networks.

While dynamic network models, such as TERGMs (Hanneke *et al.*, 2010), STERGMs (Krivitsky and Handcock, 2014), and VCERGMs (Lee *et al.*, 2020), offer valuable insights into temporal dependencies and network evolution, this study focuses on static models to address the specific ob-

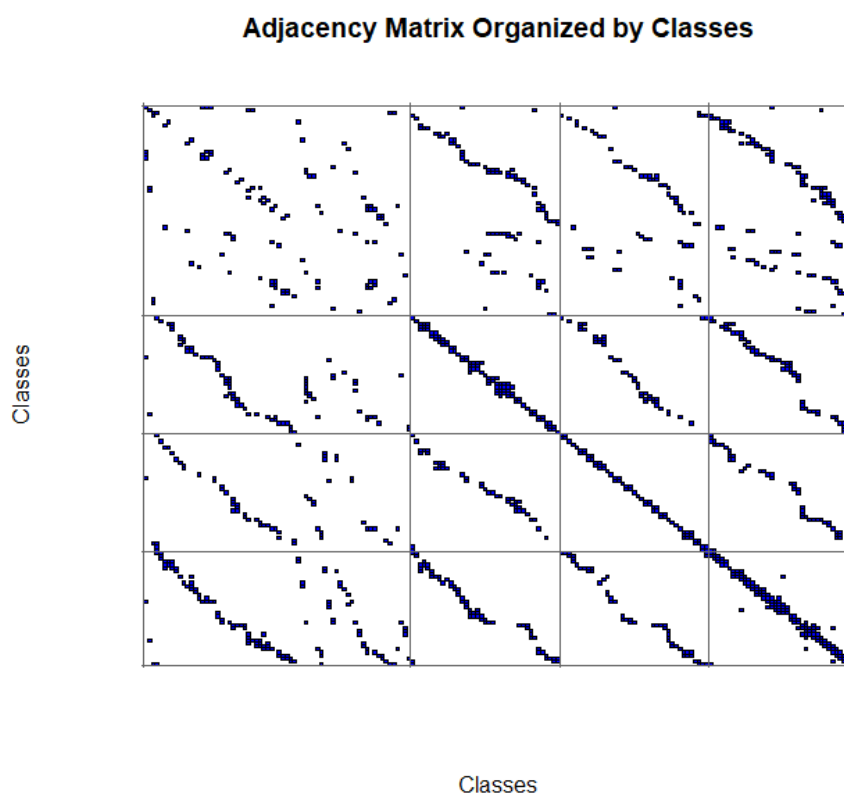


Figure 6: Adjacency matrix ordered by country classes for 2018.

jective of assessing whether the COVID-19 pandemic caused structural shifts in trade networks. By independently analyzing networks for 2018, 2020, and 2022, we were able to isolate and compare structural and covariate-driven characteristics at distinct points in time, avoiding the complexities and assumptions required by temporal dependencies. Dynamic models assume gradual transitions in network structure, which may not align with the abrupt disruptions and recovery phases observed during the pandemic. Furthermore, challenges such as high-dimensional data, sparsity in connections, and heterogeneous trade behaviors among countries make the estimation and convergence of dynamic models particularly challenging for datasets like ours. Nevertheless, dynamic approaches could provide deeper insights into temporal transitions and shifts in covariate effects in future studies, complementing the findings presented here.

The selection of 2018, 2020, and 2022 as time points was intentional and based on their ability to represent distinct phases of the pandemic's impact on international trade networks. The year 2018 serves as a pre-pandemic baseline, capturing trade dynamics under normal conditions. The year 2020 represents the peak of the pandemic, characterized by global supply chain disruptions, trade restrictions, and widespread economic shocks. Finally, 2022 reflects the recovery phase, where trade networks began to stabilize. While a more granular analysis with additional intermediate time points might offer a finer temporal resolution, this study prioritized these three representative phases to pro-

vide clear comparisons and consistent insights. The availability of comprehensive and comparable data for these years further influenced this selection. Future research with higher temporal resolution could explore the progression of trade disruptions and recoveries in greater detail, offering richer insights into the temporal nature of the pandemic's effects on trade.

The SBM results provided further insights into the structure of the international trade network. The network was divided into four classes, reflecting countries' export capacities and their roles in the global trade system. Central hubs with high connectivity were distinguished from less integrated countries, highlighting the network's hierarchical nature. A notable feature of the SBM results is the tendency of countries to trade with both structurally similar (within-class) and different (across-class) countries, reinforcing the importance of trade diversity in the global network. Importantly, the clustering patterns remained stable across the three time points, indicating no significant structural changes in the community organization of the network during or after the pandemic.

This study contributes to the understanding of bilateral trade relationships, their determinants, and the impact of the COVID-19 pandemic on the international trade network. However, it is not without limitations. The lack of post-2022 data restricted our ability to analyze the longer-term effects of the pandemic on trade structures. Additionally, computational constraints limited the inclusion of certain structural network characteristics, such as triangles and mutuality, in the ERGM. Addressing these limitations in future studies could improve model accuracy and provide deeper insights into the structure and evolution of trade networks.

Finally, we acknowledge the potential value of latent space models in complementing our analysis. These models represent nodes in a continuous geometric space, offering a powerful framework for uncovering hidden patterns and clustering tendencies in trade networks. While not applied in this study due to its focus on discrete time points and structural comparisons, latent space models could provide additional insights into the spatial organization of trade relationships and the evolution of trade dynamics over time. Incorporating these models in future research could expand the methodological framework and reveal further nuances in the international trade network, complementing the findings presented here.

Statements and declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Received October 23, 2024; Revised January 10, 2025; Accepted February 05, 2025

